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SCHOOL OF INFORMATION COMMUNICATION TECHNOLOGY



SOICT

PROJECT II REPORT:

IT3930E

**Gaussian Mixture Source Modeling and Routing for
Flow Matching**

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Project Information

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Project description

The project studies whether a learned source distribution can improve latent Flow Matching. A diagonal Gaussian mixture model is fitted to VAE data latents, and a convolutional router is trained to approximate the mixture posterior. The router then selects mixture components used to construct source latents for the transport path. Experiments compare weighted and sampled top- k source construction, endpoint-mix and bridge router training, router capacity and regularization, GMM initialization, and resume strategies. Image quality is evaluated with FID at matched checkpoints, with repeated image-generation seeds and multiple GMM fits used to separate evaluation noise from source-model uncertainty.

The implementation reuses the DiT backbone, data pipeline, and Flow Matching path of the Shortcut Models repository. It does not optimize the shortcut bootstrapping objective. For `train_type=gmm-tide`, the desired-step input is held constant through `ignore_dt=True`, while the target is the direct velocity between a routed source latent and a data latent.

Abstract

This report analyzes experiments on Gaussian-mixture source modeling and learned routing for Flow Matching. The implementation is labeled GMM-TIDE in the repository and run records, but this Project II study uses only the Flow Matching training path rather than the full shortcut objective. The experiments were completed through 23 July 2026, and the evidence is organized by algorithm rather than execution date. The historical reference uses K16, top-2, soft 0.75, and Dirichlet-512 with a single $\text{FID}_{128} = 6.969$ measurement. The strongest controlled evidence is a 2×2 factorial over source construction (weighted versus sample_topk) and router training domain (mix versus bridge), evaluated at checkpoint 400k over three GMM randomization seeds and five image-generation seeds per checkpoint. sample_topk+mix has the best aggregate mean, 7.046, but the descriptive sample-topk effect has a 95% t -interval that crosses zero. Bridge is unstable across GMM seeds. Within-checkpoint evaluation noise (sample SD 0.019–0.043) is much smaller than variation caused by refitting the GMM/source. Router distillation accuracy, flow curvature, balanced component usage, and source scale are useful failure diagnostics but none is a reliable surrogate for FID. No method in this study has yet established a matched, repeated improvement over the historical reference.

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1 Conceptual Foundations

1.1 Images and VAE latent variables

Training directly on image pixels is expensive because an image has many spatial values and contains fine-scale details that are not equally useful to the transport model. The repository first uses a variational autoencoder (VAE) [1] to map an image I to a lower-dimensional latent tensor:

$$x_1 = E_{\text{VAE}}(I).$$

The subscript 1 denotes the data endpoint of the flow. The generative model operates in this latent space rather than in pixel space. After sampling, the final latent is mapped back to an image by $\hat{I} = D_{\text{VAE}}(\hat{x}_1)$. In this project the VAE is a representation interface; the experiments change the source distribution and the transport model, not the VAE training objective.

1.2 Gaussian Mixture Models

A single Gaussian distribution represents data with one center and one covariance structure. A Gaussian mixture model (GMM) combines K Gaussian components:

$$p_{\text{GMM}}(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k), \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1.$$

Each component has a mixture weight π_k , mean μ_k , and covariance Σ_k . The components can represent distinct regions of the VAE latent distribution. This implementation uses diagonal covariance, so every latent coordinate has its own variance but cross-coordinate covariance is omitted.

The posterior responsibility

$$q_{\text{GMM}}(k | x) = \frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x; \mu_j, \Sigma_j)}$$

answers a different question from the mixture weight. The weight π_k describes how common component k is before observing x , whereas $q_{\text{GMM}}(k | x)$ measures how well component k explains a particular latent. The GMM is fitted before Flow Matching and supplies fixed component statistics during the main experiments.

1.3 Neural routing

Computing the GMM posterior provides a teacher distribution over components. The router is a convolutional network that predicts

$$q_\phi(k | x) = \text{softmax}(f_\phi(x))_k$$

and is trained to approximate that teacher. At source construction time it ranks components, retains the top- k set, and either averages component samples with normalized weights or samples one selected component. The router is therefore a component-selection mechanism. It is not a semantic image classifier, and a better posterior-distillation score does not by itself imply better generated images.

1.4 Flow Matching

Flow Matching trains a time-dependent vector field that transports samples from a source distribution to the data distribution [2]. For a routed source latent x_0 , a data latent x_1 , and $t \in [0, 1]$, this repository uses the nearly linear path

$$x_t = [1 - (1 - \varepsilon)t]x_0 + tx_1, \quad \varepsilon = 10^{-5}.$$

Its target velocity is constant along each sampled pair:

$$v^* = x_1 - (1 - \varepsilon)x_0.$$

A neural network v_ψ learns this velocity by mean-squared regression:

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\left\| v_\psi(x_t, t, \text{conditioning}) - v^* \right\|_2^2 \right].$$

At inference time, numerical integration follows the learned field from a new source latent toward a generated data latent. The role of the GMM and router is to change how x_0 and its conditioning statistics are constructed; the regression target remains a Flow Matching target.

1.5 DiT backbone and the boundary of this project

The velocity network is a Diffusion Transformer (DiT), a Transformer that operates on patches of a latent tensor rather than image pixels [3]. In the studied path, DiT receives x_t , time t , the class label, and embeddings of the routed GMM mean and standard deviation. The component identifier itself is not passed as a separate categorical input.

Shortcut Models extend a flow model by conditioning it on a requested step size and by constructing bootstrap targets for nonzero shortcut lengths [4]. That mechanism is present in the source repository, but it is outside this Project II experiment. The `gmm-tide` mode sets `ignore_dt=True`; consequently, the desired-step input is constant and the update uses the direct Flow Matching target above. “Shortcut Models” therefore identifies the codebase from which the implementation was reused, not the title or claimed scope of this project.

1.6 Fréchet Inception Distance

Fréchet Inception Distance (FID) compares the means and covariances of Inception-network features extracted from real and generated image sets [5]. If their fitted feature distributions have statistics (m_r, C_r) and (m_g, C_g) , then

$$\text{FID} = \|m_r - m_g\|_2^2 + \text{Tr} \left(C_r + C_g - 2(C_r C_g)^{1/2} \right).$$

Lower is better under a fixed evaluation protocol. FID evaluates two image sets, not individual images, and its estimate changes with sample count and random generation seed. This is why the main comparisons use matched checkpoints, the same number of generated images, and repeated evaluation seeds.

2 Executive Summary

2.1 Reference and comparison boundary

The historical reference is `tide-k16-top2-softv0p75-s128-dir512`:

$$K = 16, \quad m = 2, \quad \tau = 0.75, \quad \alpha_{\text{Dirichlet}} = 512, \quad \text{FID}_{128} = 6.969.$$

It is a GMM–TIDE run with the same main FM/DiT backbone as the subsequent ablations. However, 6.969 is one historical single-evaluation score at an unmatched checkpoint. It is therefore a reference line, not an estimate with the same repeated-evaluation protocol used in the final factorial study.

Three leaderboards must be kept separate:

Evidence class	Best observed configuration	FID128	Step	Interpretation
Historical reference	K16/top2/soft0.75/Dirichlet-512	6.969	365.6k	One unmatched evaluation; not a repeated mean.
Exploratory single evaluation	C4, GMM seed 1	6.864	320k	Best checkpoint selected from a trajectory; selection bias is possible.
Repeated checkpoint, one GMM seed	C4, GMM seed 1	6.913	400k	Mean over five evaluation seeds, but favorable only for this GMM initialization.
Repeated checkpoint, three GMM seeds	S+M	7.046	400k	Best aggregate in the matched 2×2 factorial; still above 6.969.

2.2 Main findings

1. **Sample-topk is the most promising source change, but the evidence is not conclusive.** Across three GMM seeds, its descriptive main effect relative to weighted construction is -0.117 FID with sample SD 0.154 ; the 95% t -interval is $[-0.498, 0.265]$.
2. **Bridge is not uniformly beneficial.** Its main effect is $+0.086$ FID with sample SD 0.254 , and it changes from favorable in GMM seeds 0–1 to strongly unfavorable in seed 2.
3. **GMM randomization dominates image-generation noise.** Within a fixed checkpoint, repeated-FID SD is only 0.019 – 0.043 . Across GMM fits, both cell rankings and paired effects change materially.
4. **Phase-1 accuracy and phase-2 quality are not aligned.** Router top-1 agreement or distillation KL can improve while FID worsens. The router must approximate the GMM posterior, but that approximation is not the downstream objective.
5. **Common diagnostics do not rank image quality.** Lower curvature, higher usage entropy, source magnitude close to x_1 , and lower valid loss each fail as standalone FID predictors.
6. **The velocity field remains under-dispersed.** Across many families, predicted velocity variance is only 0.63 – 0.67 of target velocity variance. Router regularization did not remove this shared failure mode.

3 Algorithm Formulation

3.1 Diagonal GMM over data latents

Let $x_1 \in \mathbb{R}^{H \times W \times C}$ denote a VAE data latent and let $\text{vec}(x_1) \in \mathbb{R}^D$. The fitted model is a *diagonal*, not isotropic, Gaussian mixture:

$$p_{\text{GMM}}(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k), \quad \Sigma_k = \text{diag}(\sigma_{k,1}^2, \dots, \sigma_{k,D}^2).$$

An isotropic component would instead use $\Sigma_k = s_k^2 I$, one scalar variance shared by all coordinates. Diagonal covariance permits a different variance per latent coordinate, giving axis-aligned ellipsoidal rather than spherical contours, while still omitting off-diagonal correlations.

The component posterior is

$$q_{\text{GMM}}(k | x) = \frac{\pi_k \mathcal{N}(x; \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x; \mu_j, \Sigma_j)}.$$

The GMM parameters π_k, μ_k, Σ_k are learned by diagonal-GMM EM; they are arrays, not neural-network outputs.

3.2 Router and distillation objective

The neural router predicts

$$q_\phi(k | x) = \text{softmax}(f_\phi(x))_k.$$

The current architecture is a convolutional trunk followed by global average pooling and a two-layer MLP:

$$\begin{aligned} x &\longrightarrow [\text{Conv}_{3 \times 3} \rightarrow \text{Norm?} \rightarrow \text{SiLU} \rightarrow \text{Dropout?}]^d \\ &\longrightarrow \text{GlobalAvgPool}_{H,W} \rightarrow \text{Dense}_M \rightarrow \text{Norm?} \rightarrow \text{SiLU} \rightarrow \text{Dropout?} \rightarrow \text{Dense}_K. \end{aligned}$$

Global average pooling averages spatial positions within each feature channel; it does not average convolution kernels. It trades spatial detail for a fixed size, translation-tolerant representation.

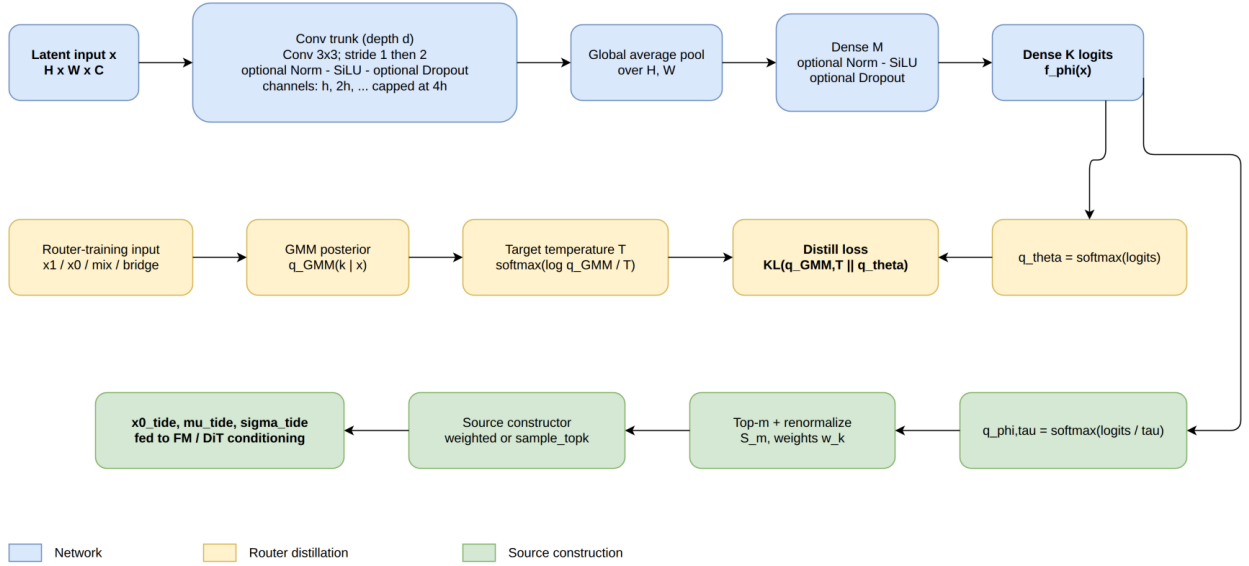


Figure 1: Router architecture and the two uses of its logits. The yellow path is GMM-posterior distillation. The green path applies source temperature, top- m routing, and source construction before FM.

For soft-KL distillation, target temperature T is applied to the *teacher target*:

$$\tilde{q}_{\text{GMM},T}(k | x) = \frac{q_{\text{GMM}}(k | x)^{1/T}}{\sum_j q_{\text{GMM}}(j | x)^{1/T}}, \quad \mathcal{L}_{\text{distill}} = D_{\text{KL}}(\tilde{q}_{\text{GMM},T} || q_\phi).$$

Thus $T > 1$ softens the labels used to train the router. This is distinct from the source temperature τ , introduced below.

3.3 Router training domains: data, mix, and bridge

The router can be distilled on several input domains:

$$\begin{aligned} x_1\text{-only: } & x^r = x_1, \\ \text{mix: } & x^r = \begin{cases} x_1, & b = 1 \\ x_0, & b = 0, \end{cases} \quad b \sim \text{Bernoulli}(p), \\ \text{bridge: } & x^r = \lambda x_1 + (1 - \lambda)x_0, \quad \lambda \sim \text{Beta}(a, b). \end{aligned}$$

The direct bridge target used here is

$$\mathcal{L}_{\text{bridge}} = D_{\text{KL}}(q_{\text{GMM}}(\cdot | x^r) || q_\phi(\cdot | x^r)).$$

Mix exposes the router to two discrete endpoint domains. Bridge fills the continuous segment between them and therefore regularizes the input-domain transition.

Bridge itself does **not** create an FM gradient path through the router. It adds distillation supervision at midpoint inputs. If the router is frozen during FM, bridge affects phase 2 only through the router learned in phase 1. If joint updates are enabled, the distillation terms update the router; an FM gradient reaches routing logits only when the route estimator is differentiable or uses a surrogate such as straight-through or Gumbel-softmax.

3.4 Source construction

First sample a base component and latent:

$$k_0 \sim \text{Categorical}(\pi), \quad x_0^{\text{base}} \sim \mathcal{N}(\mu_{k_0}, \Sigma_{k_0}).$$

The router produces source probabilities

$$q_{\phi, \tau}(k | x_0^{\text{base}}) = \text{softmax}\left(\frac{f_{\phi}(x_0^{\text{base}})}{\tau}\right)_k.$$

Here τ is the *source temperature*; it changes route sharpness immediately before top- m selection. Let

$$S_m = \text{TopK}_m(q_{\phi, \tau}), \quad w_k = \frac{q_{\phi, \tau}(k | x_0^{\text{base}})}{\sum_{j \in S_m} q_{\phi, \tau}(j | x_0^{\text{base}})} \quad (k \in S_m).$$

The main source constructors are:

$$\begin{aligned} \text{Weighted (W): } z_k &\sim \mathcal{N}(\mu_k, \Sigma_k), & x_0^{\text{tide}} &= \sum_{k \in S_m} w_k z_k, \\ \text{Sample-topk (S): } \tilde{k} &\sim \text{Categorical}(w_{S_m}), & x_0^{\text{tide}} &\sim \mathcal{N}(\mu_{\tilde{k}}, \Sigma_{\tilde{k}}), \\ \text{Hard top-1: } \hat{k} &= \arg \max_k q_{\phi, \tau}(k | x_0^{\text{base}}), & x_0^{\text{tide}} &\sim \mathcal{N}(\mu_{\hat{k}}, \Sigma_{\hat{k}}). \end{aligned}$$

Weighted construction can place the result between components. Sample-topk selects one component according to relative top- m strength and preserves that component’s covariance. It is stochastic but is not a uniform draw unless the selected probabilities are equal.

For weighted construction, the conditioning moments used by the implementation are

$$\mu^{\text{tide}} = \sum_{k \in S_m} w_k \mu_k, \quad (\sigma^{\text{tide}})^2 = \sum_{k \in S_m} (w_k \sigma_k)^2,$$

where operations on σ_k are coordinate-wise.

3.5 Straight-through and Gumbel routing

A hard top- m operation is non-differentiable with respect to component membership. The straight-through estimator uses

$$w_{\text{ST}} = q_{\phi} + \text{stopgrad}(w_{\text{hard}} - q_{\phi}).$$

The forward pass equals w_{hard} , while the backward pass uses the gradient of q_{ϕ} . Gumbel-softmax adds sampled Gumbel noise:

$$y_k = \frac{\exp((\ell_k + g_k)/\tau_g)}{\sum_j \exp((\ell_j + g_j)/\tau_g)}, \quad g_k \sim \text{Gumbel}(0, 1),$$

and can be combined with a straight-through hard sample. These methods allow an approximate FM gradient to routing logits, but they also introduce bias or variance. Their empirical results did not show an advantage in the tested regime.

3.6 Flow Matching objective

FM uses the pair (x_0^{tide}, x_1) :

$$x_t = [1 - (1 - \varepsilon)t] x_0^{\text{tide}} + t x_1, \quad v^* = x_1 - (1 - \varepsilon) x_0^{\text{tide}}.$$

The DiT predicts

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\left\| v_{\psi}(x_t, t, \Delta t, y, \mu^{\text{tide}}, \sigma^{\text{tide}}) - v^* \right\|_2^2 \right].$$

The GMM-derived μ^{tide} and σ^{tide} condition the DiT; they are not replacements for x_t .

3.7 C0–C5 candidate recipes

All C0–C5 candidates use K16/top2/soft0.75/Dirichlet-512 and the same DiT/FM backbone. The identifiers remain fixed across screening, confirmation, resume, and repeated evaluation; a later C0 or C4 is the same recipe at a different GMM seed, checkpoint, or evaluation protocol.

ID	Router data	Source	Additional change	Purpose
C0	mix (M)	weighted (W)	none	Matched control for phase-2 source experiments.
C1	bridge (B)	weighted (W)	none	Isolates bridge relative to C0.
C2	mix (M)	weighted (W)	depth 4, dropout 0.2	Isolates router capacity/regularization.
C3	bridge (B)	weighted (W)	depth 4, dropout 0.2	Tests bridge with the best local capacity change.
C4	bridge (B)	sample-topk (S)	none	Changes both the router domain and source constructor.
C5	bridge (B)	weighted (W)	PCA init + 5 Lloyd steps	Tests a different GMM initialization.

4 Regularization and Optimization Controls

4.1 Network regularization

Mechanism	Implementation and intended effect
Dropout	Applied after convolutional and MLP activations while training the router. Disabled during source generation and evaluation. It reduces co-adaptation but does not directly constrain component usage.
LayerNorm	Applied at configured convolutional/pooled/MLP locations. It normalizes each sample’s features and stabilizes activation scale. LayerNorm alone was weaker than the best dropout setting.
GroupNorm	Normalizes channel groups without batch statistics. It was tested with dropout 0.2 and 0.3 but did not improve over dropout-only.
Weight decay	Standard parameter shrinkage in the router optimizer. It regularizes weights, not posterior entropy or component balance.
EMA	Maintains $\phi_{\text{EMA}} \leftarrow \beta \phi_{\text{EMA}} + (1 - \beta) \phi$. It can stabilize a noisy jointly updated router at evaluation time. The EMA resume experiment restarted optimizer state and therefore is not a matched fresh-run comparison.

4.2 Posterior and usage regularization

Per-sample entropy floor. Let

$$h(x) = \frac{H(q_\phi(\cdot | x))}{\log K} \in [0, 1].$$

The floor penalty is

$$\mathcal{L}_{\text{floor}} = \mathbb{E}_x [\max(h_0 - h(x), 0)^2].$$

It activates only when an individual prediction is sharper than the requested entropy floor. It does not require every component to be used over a batch.

Usage prior near uniform. Let the batch-average soft usage be

$$\bar{q}_k = \frac{1}{B} \sum_{i=1}^B q_\phi(k | x_i).$$

A uniform-usage penalty can be written as

$$\mathcal{L}_{\text{usage}} = D_{\text{KL}}(\bar{q} \| u) = \sum_k \bar{q}_k \log(K \bar{q}_k) = \log K - H(\bar{q}), \quad u_k = \frac{1}{K}.$$

This term discourages global batch collapse. It can be small even when each sample is routed sharply, provided different samples choose different components.

Entropy reward. The joint objective may include

$$\mathcal{L}_{\text{entropy}} = -w_H \mathbb{E}_x H(q_\phi(\cdot | x)).$$

Unlike the usage prior, this rewards soft predictions for each sample. It can blur routing if too strong. Usage balance and per-sample entropy therefore address different collapse modes.

Tide-KL. The additional downstream-domain consistency term is

$$\mathcal{L}_{\text{tide}} = w_{\text{tide}} D_{\text{KL}}(q_{\text{GMM}}(\cdot | x_0^{\text{tide}}) \| q_\phi(\cdot | x_0^{\text{tide}})).$$

It asks the router to match the GMM posterior on generated source inputs. In the tested runs it reduced curvature but worsened FID, indicating that posterior consistency at this domain can be misaligned with image quality.

5 Metrics and Evidence Protocols

5.1 Primary and diagnostic metrics

Metric	Definition and interpretation
FID128	Primary ranking metric, evaluated with 128 denoising steps. Lower is better. FID32 is intentionally omitted from this report.
Router top-1 agreement	$\Pr[\arg \max q_\phi = \arg \max q_{\text{GMM}}] \in [0, 1]$. It measures distillation classification agreement, not source quality or FID.

Metric	Definition and interpretation
Predicted/target variance ratio	$R_v = \frac{\text{Var}(v_\psi)}{\text{Var}(v^*)}.$ $R_v < 1$ is called <i>under-dispersed relative to the FM velocity target</i>: the distribution of predicted velocity values has less variance than $v^* = x_1 - (1 - \varepsilon)x_0^{\text{tide}}$. This does not refer to a “target policy”, and it does not by itself prove mode collapse or angular error.
Flow curvature proxy	For normalized rollout segment directions $u_r = (x_{r+1} - x_r)/\ x_{r+1} - x_r\ _2$, $\kappa = \frac{1}{R-1} \sum_r \ u_{r+1} - u_r\ _2 \in [0, 2].$ Zero means consecutive segments are parallel. Lower curvature does not ensure the correct endpoint distribution.
Router usage entropy	For hard top-1 batch counts $p_k = n_k/B$, $H_{\text{use}} = -\frac{\sum_k p_k \log p_k}{\log K} \in [0, 1].$ One means uniform component use across the batch; zero means all samples select one component. This differs from per-sample posterior entropy.
Source magnitude ratio	$R_x = \frac{\mathbb{E}\ x_0^{\text{tide}}\ _2}{\mathbb{E}\ x_1\ _2}.$ Values near one rule out a first-order scale mismatch but do not establish distributional or angular alignment.
Top-k angular dispersion	$D_\angle = \sum_{k \in S_m} w_k [1 - \cos(\mu_k, \mu^{\text{tide}})] \in [0, 2].$ It measures directional disagreement among selected centers. Larger values mean more angular diversity, not necessarily better transport.
Input-domain KL	$D_{\text{KL}}(q_{\text{GMM}}(\cdot x) \ q_\phi(\cdot x))$, evaluated separately on x_0, midpoint, and x_1. “KL shift by input domain” means the router approximates the same teacher with different error depending on where x lies along the transport path.

5.2 Evidence hierarchy

Protocol	Checkpoint control	Replication	Allowed conclusion
Exploratory best checkpoint	No; best step selected	Usually one GMM fit and one eval	Candidate discovery only.
Matched 200k confirmation	Yes; same 200k horizon	One or two GMM seeds	Short-horizon source diagnostics.
Matched 400k repeated-FID	Yes; checkpoint 400k	3 GMM seeds $\times$ 5 eval seeds	Main factorial evidence.
Historical 6.969 reference	Unmatched	One reported evaluation	Context only; not a matched control estimate.

The three GMM seeds vary GMM initialization/randomization. They are not fully independent end-to-end training seeds because router/FM random-number streams were not independently resampled under a complete seed protocol. Statistical intervals over $n = 3$ must therefore be interpreted as descriptive sensitivity, not population-level uncertainty.

6 Results by Algorithm

6.1 All exploratory FID results

Figure 2 contains all 78 single-evaluation rows in the frozen plot inventory. The panels are separated by the training step where each best score was recorded, so a 180k run is not visually ranked as though it shared the 400k protocol. The dense style is retained to expose the full search surface; no causal claim is drawn from this figure.

Exploratory FID landscape, separated by training-step window

14

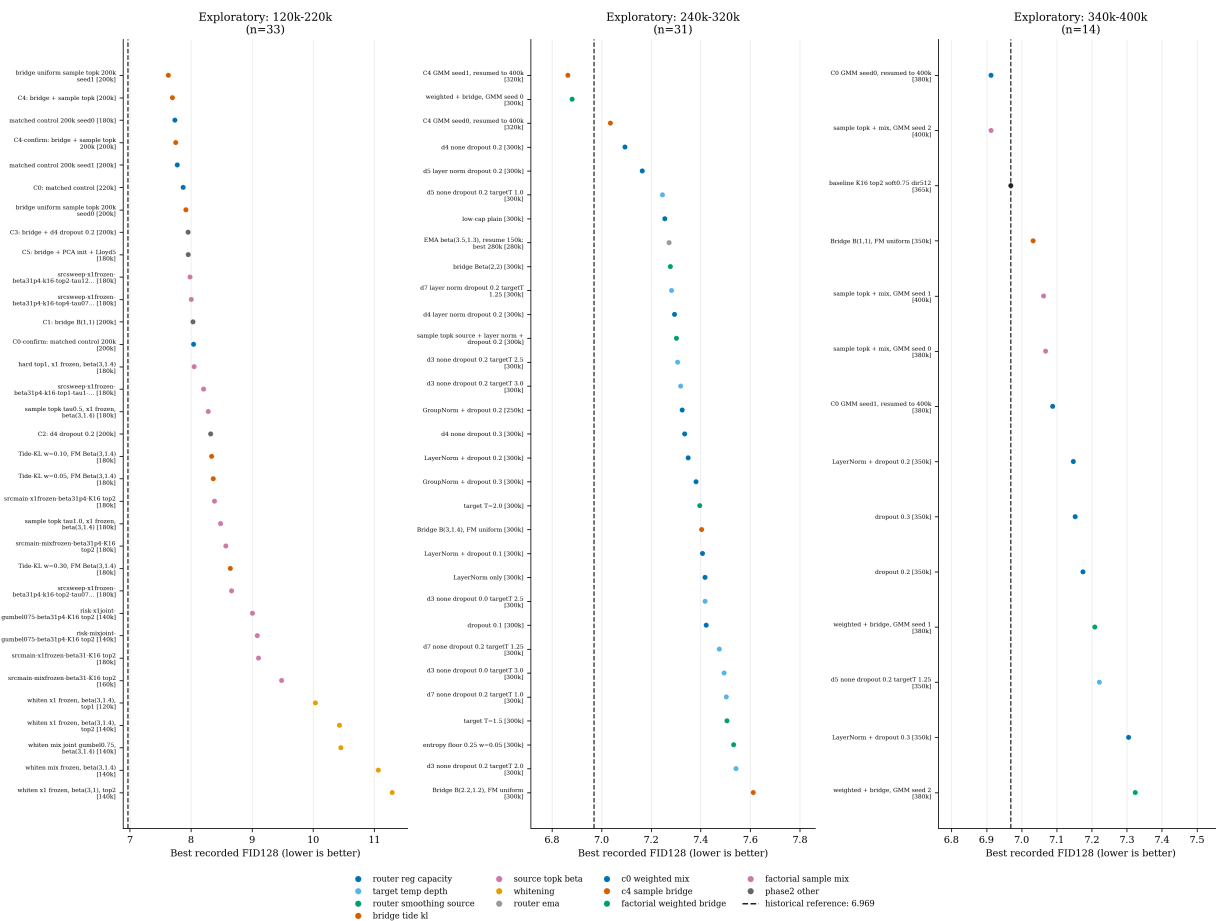


Figure 2: Exploratory FID landscape for all 78 single-evaluation records, separated into training-step windows. Point color denotes algorithm family. The dashed line is the unmatched historical reference.

6.2 Router capacity, normalization, dropout, and temperature

The best local capacity/regularization setting was depth 4 with dropout 0.2. Increasing depth to 5 or 7 did not continue the trend. Target temperature softened distillation labels but did not create a monotonic FID improvement.

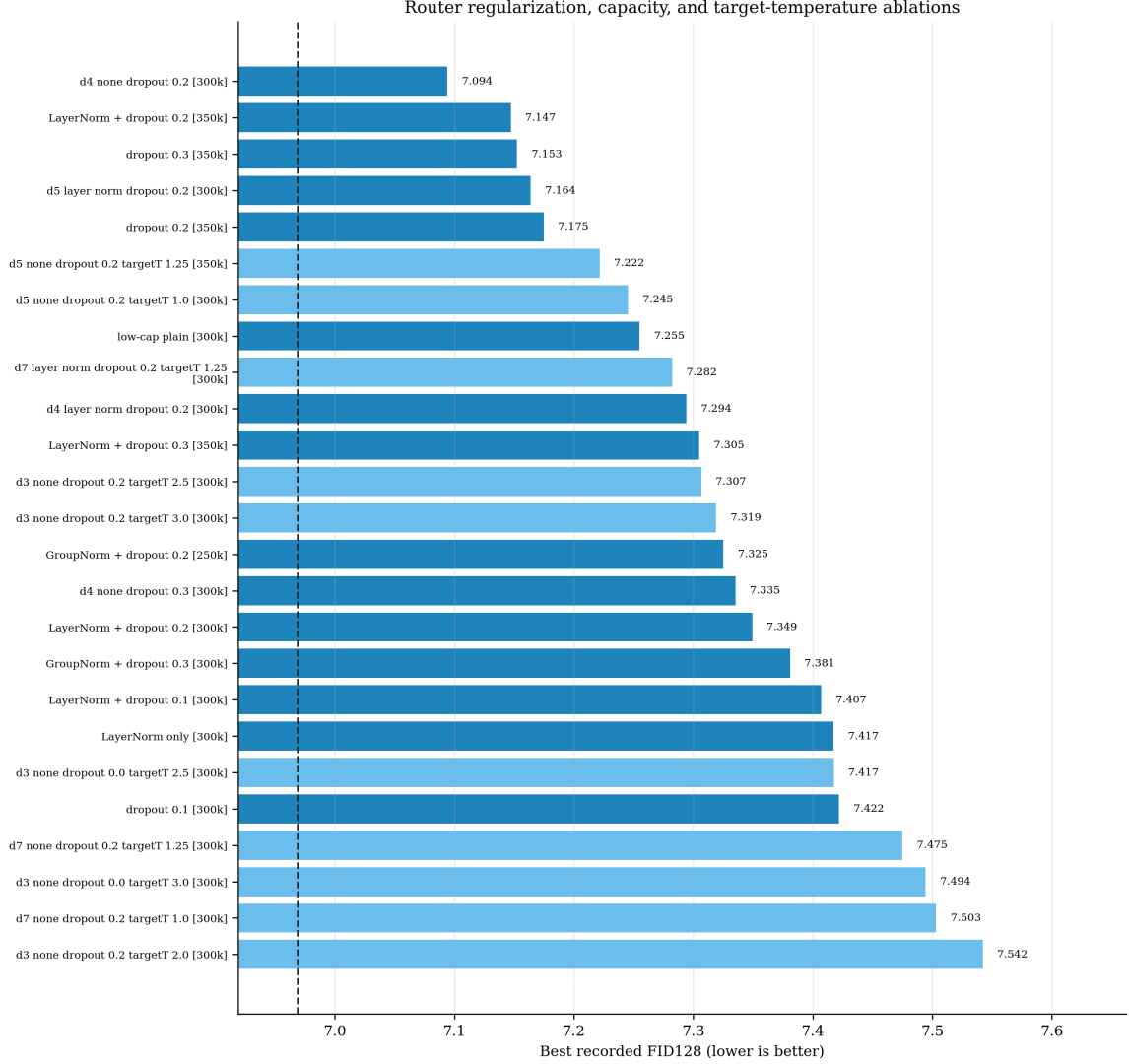


Figure 3: Router regularization, capacity, and target-temperature ablations. Values are exploratory best single-evaluation scores; the checkpoint step is shown in each label.

Configuration	Depth	FID128	Step	Top1	R_v	Curv.	Usage H
dropout 0.2, no norm	4	7.094	300k	0.943	0.651	0.02084	0.949
LayerNorm + dropout 0.2	3	7.147	350k	0.912	0.669	0.02082	0.944
dropout 0.3, no norm	3	7.153	350k	0.914	0.628	0.02083	0.950
LayerNorm + dropout 0.2	5	7.164	300k	0.964	0.661	0.02091	0.948
dropout 0.2, target $T = 1.25$	5	7.222	350k	0.961	0.635	0.02080	0.950
low capacity, plain	2	7.255	300k	0.820	0.674	0.02072	0.940
LayerNorm + dropout 0.2	7	7.282	300k	0.936	0.643	0.02087	0.955
GroupNorm + dropout 0.2	3	7.325	250k	0.914	0.633	0.02068	0.946
LayerNorm only	3	7.417	300k	0.907	0.650	0.02060	0.947
dropout 0.1, no norm	3	7.422	300k	0.915	0.662	0.02071	0.945

The depth-5 model attains better router top-1 agreement than the depth-4 winner but worse FID. This is direct evidence that router imitation accuracy is not a downstream ranking metric.

6.3 Router smoothing and bridge inputs

Configuration	FID128	Step	Top1	R_v	Curv.	Usage H
bridge $B(1, 1)$, FM uniform	7.033	350k	0.907	0.672	0.02093	0.884
bridge $B(2, 2)$	7.277	300k	–	0.650	0.02088	0.939
sample-topk + LayerNorm + dropout 0.2	7.302	300k	–	0.656	0.02069	0.952
target $T = 2.0$	7.396	300k	–	0.643	0.02086	0.939
bridge $B(3, 1.4)$, FM uniform	7.404	300k	0.868	0.665	0.02062	0.864
target $T = 1.5$	7.506	300k	–	0.644	0.02077	0.947
entropy floor 0.25, weight 0.05	7.533	300k	–	0.649	0.02069	0.946
bridge $B(2.2, 1.2)$, FM uniform	7.612	300k	0.888	0.660	0.02080	0.896

Uniform bridge was the best exploratory router-domain change. Endpoint-biased bridge distributions were worse, suggesting that broad path coverage mattered more than concentrating router samples near x_1 . This observation did not survive all GMM seeds in the matched factorial, so it remains conditional.

6.4 Source routing, Gumbel/ST, whitening, and Tide-KL

Configuration	FID128	Step	Top1	R_v	Curv.	Usage H
x_1 -frozen, top2 $\tau = 1.25$, beta(3,1.4)	7.986	180k	–	0.637	–	–
x_1 -frozen, top4 $\tau = 0.75$, beta(3,1.4)	8.006	180k	–	0.635	–	–
hard top1, beta(3,1.4)	8.053	180k	1.000	0.644	0.01848	0.908
sample-topk $\tau = 0.5$, beta(3,1.4)	8.284	180k	1.000	0.634	0.01860	0.920
Tide-KL weight 0.10	8.339	180k	0.900	0.662	0.01903	0.947
Tide-KL weight 0.05	8.366	180k	0.917	0.659	0.01867	0.926
sample-topk $\tau = 1.0$, beta(3,1.4)	8.484	180k	1.000	0.640	0.01885	0.935
x_1 -joint Gumbel-ST 0.75	9.007	140k	–	0.661	–	–
mix-joint Gumbel-ST 0.75	9.086	140k	–	0.657	–	–
whiten x_1 -frozen, top1	10.037	120k	1.000	0.686	0.01867	0.970
whiten x_1 -frozen, top2	10.431	140k	1.000	0.675	0.01892	0.972
whiten mix-joint Gumbel-ST	10.453	140k	1.000	0.641	0.01872	0.936

Whitening decorrelated or rescaled latent channels for GMM fitting/source use, then inverted the transformation before FM. It tested whether Euclidean distance in the raw latent coordinates distorted component geometry. Whitening produced balanced usage and reasonable variance ratios but substantially worse FID in this implementation. Gumbel/ST also failed to improve the early trajectory. Tide-KL lowered curvature while worsening FID, showing that a straighter rollout can still terminate at a worse distribution.

6.5 EMA resume

The beta(3.5,1.3) EMA experiment resumed model/router parameters from step 150k but discarded old optimizer state and initialized a new optimizer. Its FID improved from 8.308 at 160k to 7.272 at 280k, then drifted to 7.361 at 300k and 8.758 at 340k. Because pre-150k history and optimizer state differ from a fresh run, this result is evidence about checkpoint continuation, not a fair algorithm comparison.

6.6 Matched C0/C4 confirmation at 200k

The first clean comparison held the horizon at 200k and used one GMM initialization. C4 was better than C0 at all ten evaluation points.



Figure 4: Matched 200k clean confirmation. C4 is better throughout this trajectory, but the experiment contains only one GMM initialization.

Repeating both recipes over GMM seeds 0 and 1 weakened that conclusion: C0 had best-FID mean 7.757 with SD 0.019, while C4 had mean 7.775 with SD 0.143. Figure 5 uses only metrics logged at the same 200k checkpoint. It does not pair a best earlier FID with last-step diagnostics.

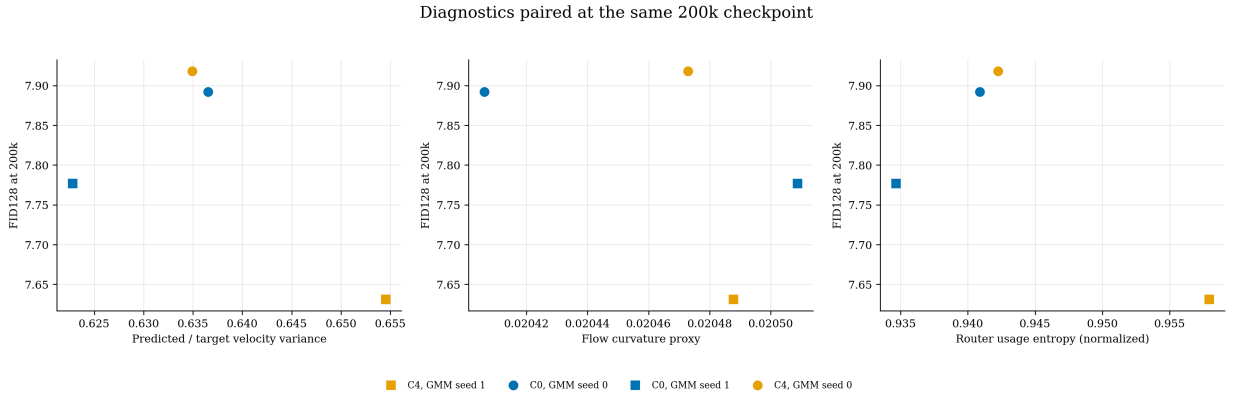


Figure 5: Four matched 200k runs. FID, variance ratio, curvature, and usage entropy all come from the same checkpoint. None of the three diagnostics orders the four FID scores consistently.

6.7 Matched 2x2 factorial at 400k

The final factorial separates two factors:

source constructor: $W = \text{weighted}$, $S = \text{sample_topk}$,
 router data domain: $M = \text{mix}$, $B = \text{bridge}$.

C0 is $W+M$ and C4 is $S+B$. The missing cells $W+B$ and $S+M$ are required to attribute effects.

Every cell was evaluated at checkpoint 400k for GMM seeds 0, 1, and 2. Every checkpoint used evaluation seeds $\{101, 202, 303, 404, 505\}$, with 50,048 generated samples per evaluation seed. This gives 12 checkpoint means and 60 FID measurements.

GMM seed	W+M	W+B	S+M	S+B	Best cell
0	7.075	7.098	7.170	7.040	S+B
1	7.266	7.252	7.038	6.913	S+B
2	7.102	7.394	6.931	7.397	S+M
Mean	7.147	7.248	7.046	7.117	S+M
SD across GMM seeds	0.104	0.148	0.120	0.251	–

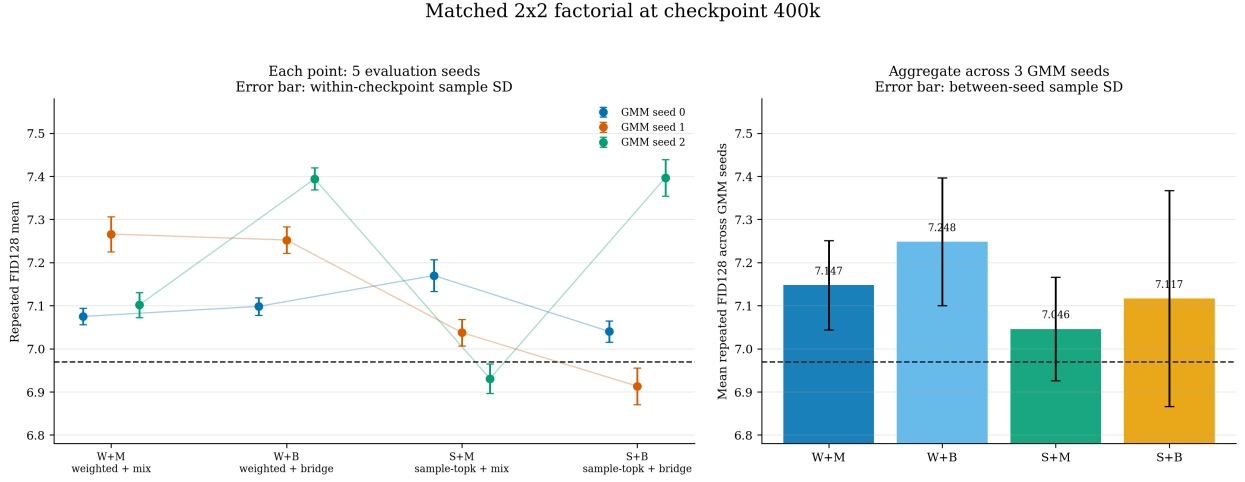


Figure 6: Repeated FID for the complete factorial at checkpoint 400k. The left panel separates within-checkpoint evaluation noise from GMM-seed variation. The right panel aggregates across the three GMM seeds.

The descriptive effects, using negative values to favor sample-topk or bridge, are:

$$\begin{aligned}\Delta_{\text{source}} &= \frac{1}{2}[(S+M) + (S+B) - (W+M) - (W+B)] = -0.117, \\ \Delta_{\text{router}} &= \frac{1}{2}[(W+B) + (S+B) - (W+M) - (S+M)] = +0.086, \\ \Delta_{\text{interaction}} &= (S+B - S-M) - (W+B - W-M) = -0.030.\end{aligned}$$

Across the three GMM seeds, their sample SD values are 0.154, 0.254, and 0.178, respectively. With $n = 3$, the corresponding 95% t -intervals are approximately

$$[-0.498, 0.265], \quad [-0.546, 0.718], \quad [-0.471, 0.411].$$

All cross zero.

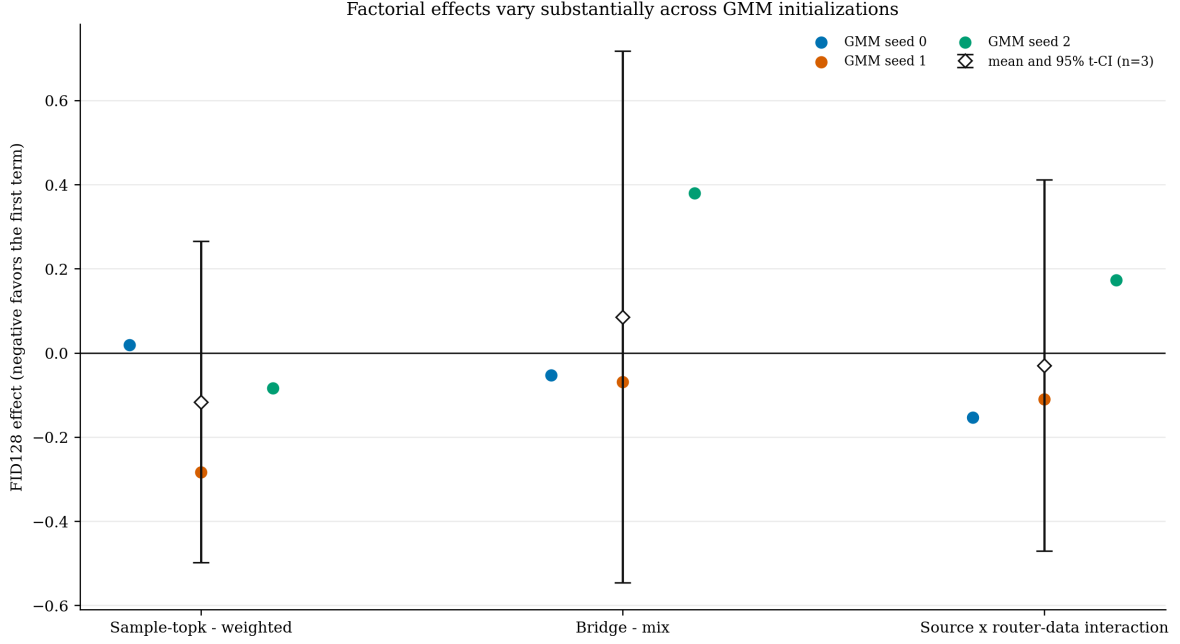


Figure 7: Factorial effects by GMM seed and descriptive 95% t -intervals over the three initializations. Sample-topk has the most favorable mean, but no effect is stable enough for a causal claim.

The cell with the best mean is S+M, not the bridge C4 recipe. This resolves an earlier apparent contradiction: bridge $B(1, 1)$ had the best exploratory single score among router-domain ablations, while the matched repeated factorial favors sample-topk with mix on average.

6.8 Geometry and router-domain audit

The GMM-seed-2 audit compares C0 and C4 at the seed where C4 loses by 0.295 repeated FID. Aggregate source geometry is nearly identical:

	C0	C4
x_0/x_1 magnitude ratio	1.0050	1.0058
$\cos(x_0, x_1)$	0.1742	0.1744
covariance condition number	3.385	3.380.

These scalars cannot explain the FID gap.

The top-k angular dispersion is more different:

$$D_{\angle}(C0) = 1.16 \times 10^{-4}, \quad D_{\angle}(C4) = 1.34 \times 10^{-3}.$$

C4 retains more directional diversity among selected centers, but greater angular diversity does not guarantee better endpoint transport.

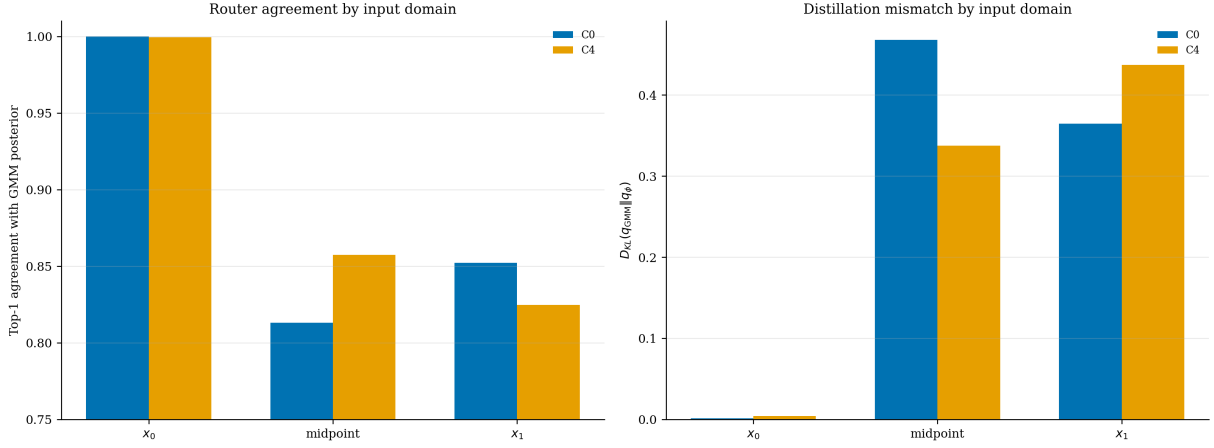


Figure 8: Input-domain router audit at GMM seed 2. C4 matches the teacher better at midpoint but worse at x_1 . The router can therefore improve on the bridge domain while degrading at the data endpoint.

“Input-domain KL shift” refers to this change in approximation error across x_0 , midpoint, and x_1 . It is not a different KL formula. The same $D_{KL}(q_{GMM} || q_\phi)$ is evaluated on three input distributions. Bridge shifts training mass toward midpoints, so improved midpoint KL can coexist with worse endpoint KL.

7 Interpretation

7.1 Phase-1 and phase-2 objectives are misaligned

Phase 1 minimizes a posterior imitation objective:

$$\min_{\phi} \mathbb{E}_{x \sim \mathcal{D}_{\text{router}}} D_{KL}(q_{GMM}(\cdot | x) || q_{\phi}(\cdot | x)).$$

Phase 2 needs the induced source distribution and velocity field to produce a good endpoint distribution:

$$\min_{\psi, \text{possibly } \phi} \mathbb{E} \|v_{\psi} - v^*(x_0^{\text{tide}}(\phi), x_1)\|^2, \quad \text{final selection by FID.}$$

The objectives differ in both input domain and consequence. A router can imitate the GMM posterior accurately but select or combine components in a way that is poor for downstream transport. Conversely, a router with weaker top-1 agreement can induce a source that is easier for FM.

The experiments support this mismatch:

- depth 5 has stronger router agreement than the depth-4 FID winner;
- C4 clean-confirm has worse router validation metrics than C0 but better FID at 200k;
- Tide-KL improves a downstream posterior-consistency diagnostic and curvature while worsening FID;
- bridge improves midpoint KL at GMM seed 2 but worsens x_1 KL and loses in FID.

7.2 What the factorial establishes

The factorial removes the confounding in C0 versus C4. Its strongest descriptive signal is that sampling one selected component is preferable to averaging samples from multiple components. This is consistent with the geometric concern that weighted averaging can create points between component supports. However, GMM seed 0 contradicts the average direction and the $n = 3$ interval crosses zero. The appropriate conclusion is *sample-topk is the leading candidate*, not *sample-topk is proven superior*.

Bridge is more fragile. It helps for some source/seed combinations and hurts strongly for others. The audit suggests that midpoint supervision can trade off against endpoint fidelity. Bridge should therefore be treated as an interaction with source and GMM fit, not as a generally beneficial router augmentation.

7.3 Why diagnostics do not replace FID

Each diagnostic removes one simple failure explanation:

- $R_x \approx 1$ rules out a gross norm mismatch.
- High H_{use} rules out global one-component collapse.
- Low curvature indicates locally straighter rollout segments.
- High top-1 agreement indicates faithful posterior classification.
- Low domain KL indicates local teacher imitation.

None verifies the final generated distribution. FID depends on the entire transport map, the shape of each conditioned source component, and decoder output. Consequently, these metrics should be used to diagnose *why* a run fails, not to select a winner without image-distribution evaluation.

7.4 Uncertainty structure

There are at least three uncertainty levels:

1. evaluation seed noise for a fixed checkpoint;
2. GMM/source randomization, including the posterior targets seen by the router;
3. full optimization randomness in router and FM training.

The study estimates the first level and partially probes the second. It does not independently replicate the third. The observed ranking reversals across GMM seeds show that reporting only a best checkpoint or only repeated image seeds is insufficient.

8 Compute and Reproducibility

8.1 Approximate resource requirement

The operational training budget is constrained by an approximately 8.75–9.0 hour TPU session. A 400k pipeline therefore requires an initial run and one resume session. For the complete factorial:

Quantity	Value	Scope
Full training pipelines	12	4 cells $\times$ 3 GMM seeds.
TPU sessions	24	Approximately 2 sessions per 400k pipeline.
TPU allocation-hours	210–216	$24 \times (8.75\text{--}9.0)$; excludes exploratory sweeps and failed attempts.
Repeated-FID checkpoint evaluations	12	One per cell and GMM seed.
FID evaluation seeds	60	12×5 .
Generated samples for repeated FID	3,002,880	$60 \times 50,048$.

“TPU allocation-hours” refers to wall-clock hours for the allocated v5e-8 session, not the sum of individual chip-core hours. The estimate intentionally excludes heterogeneous exploratory sweeps, retries, and infrastructure failures because their runtime accounting is not uniform.

8.2 Reproducibility contract

- Runtime code and experiment protocols are fixed at commit 9fa1ea9.
- The original pre-revision report snapshot is commit 442ad5a on branch document.
- Plot generation is reproducible with `uvrunpythonscripts/generate_algorithm_report_plots.py`.
- The generated numerical snapshot is `pdf/figures/algorithm_report_data.json`.
- Repeated-FID uses checkpoint 400k, seeds $\{101, 202, 303, 404, 505\}$, and 50,048 generations per seed.

- Model evidence is taken from parsed JSON/JSONL/CSV diagnostics or W&B histories. Checkpoints were not downloaded for report analysis.
- The historical 6.969 score is explicitly labeled as unmatched single-evaluation evidence.

9 Limitations and Conclusions

9.1 Limitations

- Only three GMM randomization seeds are available for the complete factorial.
- These are not three fully independent router/FM training seeds.
- The historical reference lacks matched repeated evaluation.
- Exploratory runs stop at different steps; Figure 2 is descriptive and separated by step window for this reason.
- Several diagnostics are unavailable for older runs. Missing entries are not imputed from family averages.
- Diagonal GMM covariance cannot model cross-coordinate correlations.
- FID is the only downstream image-distribution ranker in this report; diagnostics do not establish perceptual equivalence.

9.2 Conclusions

The experiments do not yet demonstrate a robust improvement over the historical GMM-TIDE reference. They do narrow the technical problem. Sample-topk is the strongest source-construction candidate, while bridge is seed- and source-dependent. Router overfitting in the narrow supervised sense is not the sole bottleneck: common regularizers improve some router metrics without a corresponding FID gain. The decisive unresolved issue is alignment between GMM-posterior imitation in phase 1 and transport quality in phase 2. Future claims should be gated by matched checkpoints, repeated image seeds, multiple GMM fits, and independent end-to-end optimization seeds.

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- [1] D. P. Kingma and M. Welling, “Auto-Encoding Variational Bayes,” in *International Conference on Learning Representations*, 2014. arXiv:1312.6114.
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- [3] W. Peebles and S. Xie, “Scalable Diffusion Models with Transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023. arXiv:2212.09748.
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A Local Evidence Sources

- reports/gmm_tide_fm_beta_source10_data_available_20260609.json
- reports/gmm_tide_router_all_combined_20260611.json
- reports/bridge_tidekl_analysis_20260612.json
- reports/router_ema_beta35_resume_diagnostics_20260613.json

- reports/gmm_tide_phase2_aware_c0_c5_analysis_20260623.json
- reports/gmm_tide_phase2_aware_confirm_c0_c4_200k_analysis_20260629.json
- reports/gmm_tide_phase2_variance_repeats4_analysis_20260707.json
- reports/gmm_tide_phase2_variance_resume2_420k_results_20260710.json
- reports/gmm_tide_fid_repeat_audit_complete_20260715.json
- reports/gmm_tide_fid_confirmation_gmmseed2_result_20260718.json
- reports/gmm_tide_factorial_resume400_retry2_analysis_20260720.json
- reports/gmm_tide_factorial_wb_sm_fidrepeat6_results_20260723.json
- reports/progress_report_evidence_20260721.json
- pdf/figures/weekly_plot_data.json